

Visioball



Project by:
Diede Wip

Fontys
&

Visio

Abstract

This project focuses on enhancing the Visioball, a specialized ball designed for visually impaired children. The primary goals are to evaluate the previous version of the ball (Visioball 1.0) and to develop an improved proof of concept and functional prototype. The project is conducted in several phases, starting with the evaluation of the existing design, followed by conceptualization and design, prototyping, and final evaluation.

Key findings from the evaluation revealed several issues with the previous design, including excessive weight, inconsistent sound production, and structural problems. To address these, new design requirements were established, focusing on weight reduction, sound consistency, and durability. Three concepts were explored: analogue, electronic, and digital. The digital concept, utilizing a gyroscope and microcontroller, was identified as the most promising due to its potential for customization and robust performance.

The improved Visioball aims to provide a better user experience for visually impaired children, ensuring the ball is easier to locate when stationary, rolls straight, and produces consistent sound. This project highlights the importance of iterative design and user-centric approaches in developing assistive technologies.

Contents

Abstract.....	2
Project plan	5
Introduction	5
Problem definition	5
Project goals.....	5
General goals:	5
Personal goals:	5
Assignment.....	5
Approach.....	5
Deliverables.....	6
Planning	6
Milestones:.....	6
Evaluation Visioball 1.0	7
Autopsy	7
Functional tests	8
Straightness of roll:	8
DB test:.....	9
Consistence of sound:	9
Problems & solutions	10
Requirements.....	11
Concepts	12
Analogue	12
Electronic	13
Digital	15
Conclusion.....	15
Research.....	16
Ideation closing mechanism.....	18
Evaluation closing.....	19
Requirements closing mechanism	19
Kesselring	20
Prototyping	21
Electronic inner ball	21
Proof of principle	23
Prototype	24
Feedback	27

First prototype outer ball	28
<i>Main goals:</i>	28
InsertsV1	29
InsertsV2	30
Feedback proto-1:	31
<i>Mould:</i>	31
<i>Design:</i>	31
<i>Inserts:</i>	31
<i>Loose Feedback:</i>	31
Prototype 2	32
Main goals:	32
Requirements design	32
Requirements mould.....	33
Requirements inserts	33
Appendix	34
Autopsy components list.....	34
Functional tests	35
Final code prototype	37
Instructions:	37
Technical Drawings.....	39

Project plan

Introduction

Koninklijke Visio is an organization that helps visually impaired people. Visio offers information, advice, various forms of examination, counselling, rehabilitation, education, and housing to visually impaired people and to people with mental, physical, or other sensory disabilities. One branch of Visio consists of multiple specialized schools in the Netherlands.

Problem definition

When playing ball games with blind children a ball with bells is used. However, when this ball is not in motion the children have difficulty finding them, due to the lack of sound produced by the ball.

Project goals

General goals:

- Evaluating the previous visioball.
- Making an improved proof of principle.
- Translating the proofs of principles into a proof of concept.

Personal goals:

- Testing production techniques new to me.
- Creating and documenting the design process.
- Learning about how other (visually impaired) people see and experience the world.

Assignment

Evaluate the current proof of principle. Use the information learned by this to get a better understanding of the want and needs of all the user groups. To eventually design a better fitting concept and prototype. While keeping the feasibility and production in mind.

Approach

Phase 1, evaluating and testing:

- Project plan
- Evaluation last year's project
- Testing last year's project

Phase 2, Conceptualising and designing:

- Research & orientation.
- Dividing the project into concrete parts.
- Finding solutions / designs for each part.

Phase 3, Prototyping & evaluating:

- Electronics and software set-up.
- Physical model.
- Bluetooth / app (Only if necessary).
- Complete prototype.
- Evaluation prototype and project.

Deliverables

- Proof of principle.
- Proof of concept.
- Functional prototype.
- Documentation of design process and design rationale.
- Technical documentation (i.e. drawings, material specifications, production techniques, CAD, etc).

Planning

[illegible]

Milestones:

- Ideation
- Proof of principle
- Proof of concept
- Prototype

Evaluation Visioball 1.0

Autopsy

For a more detailed list see the Appendix

Goal:

What components did they use, and why?

How did they construct it, and why?

Outside ball:

The outside consists of an existing foam ball, cut open and glued together.

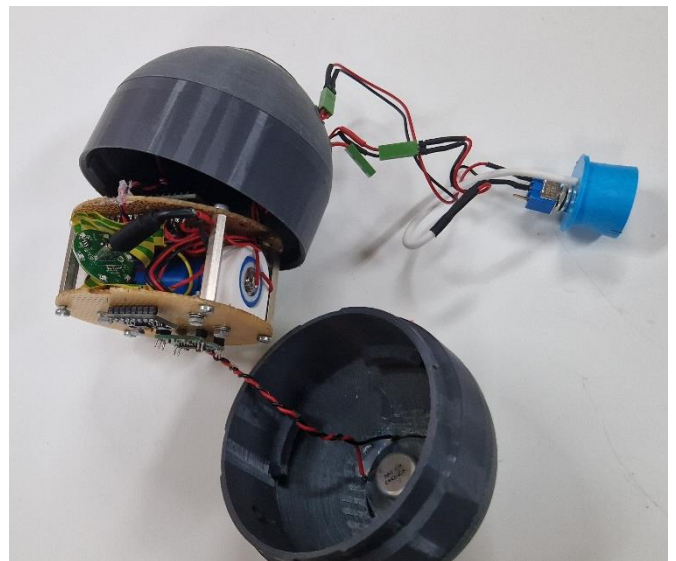
Between the outside and the inside ball, we find a layer of foam to dampen the blows and hold the inside ball in place.

Inside ball:

3D-printed hollow capsule that twist locks, with 2 small holes for cables to pass through and 2 large holes for speakers to be seated in.

Electronics:

- Speaker x2
- Amplifier
- Arduino Nano
- SD card reader
- Gyroscope
- Rechargeable battery
- Charger
- Switch
- 9V battery



Oddly enough the 9V battery seems to be connected to nothing at all adding 35 unnecessary grams.

Coding:

Within the code the first read the X, Y, & Z values and then calculate the difference within a time interval.

```
1. X_difference = abs(X_out) - abs(X_out2);
2. X_difference = abs(X_difference);
3. Y_difference = abs(Y_out) - abs(Y_out2);
4. Y_difference = abs(Y_difference);
5. Z_difference = abs(Z_out) - abs(Z_out2);
6. Z_difference = abs(Z_difference);
```

Afterwards the differences are added into Total_difference:

```
1. Total_difference = X_difference + Y_difference + Z_difference;
```

Finally, the Total_difference is used to determine the frequency of the sound fragment.

```
1. else {
2.     tmrpcm.setVolume(5);
3.     tmrpcm.play("bats.wav");
4.     interval = (700 - (Total_difference * 600));
5.     if(interval <= 0){
6.         interval = 50;
7.         State = 0;
8.     }
```

Functional tests

Only the conclusions are presented here, for the full test document see the Appendix.



Straightness of roll:

Goal:

When the CG is off the ball won't roll straight, does the ball roll straight?

The test:

Roll the ball over a straight 12m line (workshop hallway), with different starting speeds and note the deviation from the line.

Conclusion:

When the ball is rolling fast there is no noticeable deviation in its path. However, when the ball is going slow the deviation is noticeable, with deviations of up to 700mm being observed.

DB test:

Goal:

Is the ball loud enough, and is the loudness equally at all angles?

Test:

Measure the loudness in dB of the ball from 1m away, do this both for stationary and moving, as well as for speakers to the side and speakers directly facing the dB meters (dB measuring apps on 4 different phones).

Conclusion:

The lowest volume recorded was 48dB (stationary speakers facing the phones), and the highest 52.5dB (Rolling speakers to the side). The volume is overall too soft, with last year's goal being >60dB.

There doesn't appear to be a notable difference between the speaker orientation, which can be observed when rolling the ball. This might have to do with using a cocoon for the measurements, and the echo's evening out the results.

Consistence of sound:

Goal:

We observed that the sound isn't very consistent, can we quantify this and find out why?
Is there a correlation between speed and consistency?

Test:

Roll the ball over a known distance, measure the time, and count the number of sound bleeps made in that interval.

Conclusion:

The results are inconsistent, making it difficult to make conclusions.

What we can say is that the speed seems to influence the consistency of the sound, with slow being more consistent than fast.

Problems & solutions

What problems do we see in the old prototype and what can we do to improve it.

Problem	Possible solution(s)
To heavy.	- Make the core lighter. - Get rid of redundant electronics. - Mount everything on one PCB. - Change the outside of the ball to a lighter material.
Inconsistency of sound: Likely caused by how the movement is calculated within the code.	- Change how the movement is calculated within the code. - Use sensor fusion (a technique that combines multiple sensors into one giving more accurate values).
Damaged speakers.	- Make the holes in the outer ball smaller, so that nothing can enter. - Have no holes in the outer ball. - Use different speakers. - Mount them in a more hidden manner.
Ball falling apart.	- Have a proper closing mechanism. - Better glue. - Less pressure from the inside.
Ball not turning on every time: Most likely caused by one of the electronic components not getting a high enough voltage.	- Change the electronic components. - Change the charging circuitry.
Not rolling straight: Multiple problems, it isn't round, the CG is of, and there is a notable seem.	- Make the ball round. - Make sure that the CG is in the right place. - Close the ball in a manner that has no notable seem.
Sound is too soft.	- Increase the wattage to the speakers. - Have different speakers. - Increase the number of speakers. - Decrease the amount of stuff between the speakers and the outside. - Change the sound.
Unequal sound / sound not 360	- Increase the number of speakers. - Create a resonance box. - Change the path that the sound travels. - Changing the sound itself (longer waves can travel over sharper corners).

Requirements

Requirement	Type part	Goal	Demand / Wish	Priority
All the electronics must be powered on 5V.	Electronics	5V circuit, this has to do with safety and battery life.	Demand	5
Gravity needs to be correct for the ball to roll in a straight line.	User Experience	Centre of gravity can be ... mm from the centre of the ball.	Demand	4
The system has a working duration of 3+ hours with one full charge.	Electronics	(+)3 hours duration	Wish	3
The ball should produce sound when stationary.	User Experience		Demand	5
The range of sound volume should be minimum 60 dB and 85 dB is suitable.	Useability	Adjustable volume.	Demand	5
The maximum frequency of the sound should be 8000 hertz and should be audible in a gym class with a dB of: 60 (not specified if it was a gym class).	User experience	The frequencies fall within the hearable spectrum.	Demand	4
The size of the ball should be size 23 or the 16 cm.	Design	Both a small and large ball must be developed.	Demand	5
The ball gives light. The maximum lumen the ball should emit is less than 500 lumen.	User experience	Enough light to make it stand out to partially blind people, while being dim enough to prevent eye-pains.	Wish?	Check with research 3
The weight of the ball should be a maximum of 800 grams	User experience	For goalball there is a maximum weight allowed.	Demand	
The ball should survive a .. m drop test	Durability	Surviving rough play.	Demand	
The casing of the electrical components should have a .. thermal conductivity	Electronics	Prevent the electronics from overheating.	Demand	5
We design for a production of 30 to 50 balls	Production	This is the estimate given by the client.	Demand	5
The design of the ball should have contrasting colours with the playground	User experience	Contrasting colours help the partially blind children.	Demand	5
Ball should not have holes big enough for the fingers	Safety	Holes within the ball are sufficiently small to prevent fingers entering. Both for safety of the user and the electronics.	Wish	4
The surface of the ball should give enough grip.	User experience	Determine the right texture for the outside of the ball.	Wish	3
The hardness of the ball should be ...		Like Visioball foam.	Demand	5
The lifespan of the ball should be minimum 3 year.	Sustainability		Demand	4
The ball should be waterproof IP rating	Useability	The ball works in damp to moist grass.	Wish	
Controllable by phone.	User experience (teacher)	The volume, and on/off can be controlled via a phone.	Demand	4
Lights in the ball.	User experience	For partially blind children it's nice to see where the ball is.	Wish	3
Inside electronics needs to be protected.	Sustainability	The electronics can survive rough play, and the determined IP Rating.	Demand	5
Chargeable from the outside of the ball.	User experience (teacher)	The ball can be charged without having to take anything apart.	Demand	5
Should be able to be repaired doesn't have to be done by the end user.	User experience (Repairman)	The ball can be taken apart non-destructively in an intuitive manner.	Demand	5
Adjustable volume	User experience	The volume can be adjusted over a range of ...dB to ...dB.	Wish	4
Different choice melodies via app	User experience	Multiple melodies can be chosen, both for the moving and the stationary state.	Wish	3
Use a common charger	User experience	The ball uses USB-C (as determined by the EU).	Demand	4
Should be able to be picked up with one adult male hand (Small ball).	User experience	The small ball can be picked up one-handed by an average Dutch adult male.	Demand	5

Concepts

Before the first client meeting (when we set up the requirements list) I made 3 concepts based on three different functioning designs.

The goal of these concepts is to narrow down the search, to find out which of the 3 directions has the most potential.

Analogue

Summary:

Exclusively using analogue means, this means keeping the bells and figuring out a way to keep them ringing for a certain amount of time after becoming stationary.

Problem definition:

The system does not produce a sound long or loud enough after becoming stationary for the users to act and react.

Goal:

Create a system that produces a sound after becoming stationary for long and loud enough for the users to act and react.

Research:

What 'bells' can we use for long time chiming?

We could use hanging bells that keep moving even after the ball becomes stationary.

Chimes can keep making sound for >20 seconds.



End of concept!

At this point I was advised by last years group to abandon this concept, because the client has specifically stated that he does not want analogue systems. Reasons being most long sounding systems having annoying sounds, the lack of custom sounds, and even the best systems won't produce sound long enough when the ball is properly lost (i.e. bushes, or long grass.)

Electronic

Summary:

Instead of using a micro controller to read a gyroscope we could make a electronic system with mechanical gravity switches.

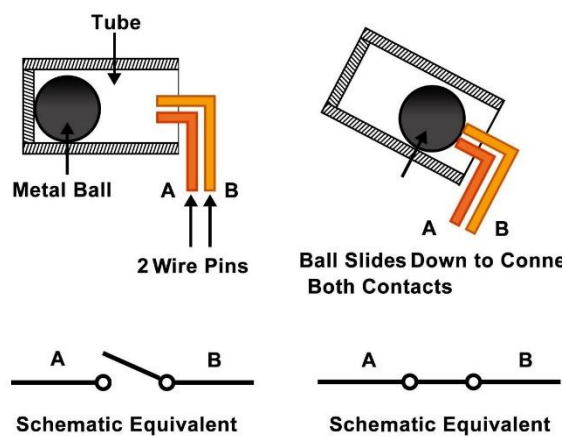
Research:

How do we measure or observe movement?

Tilt switches, gravity switches.

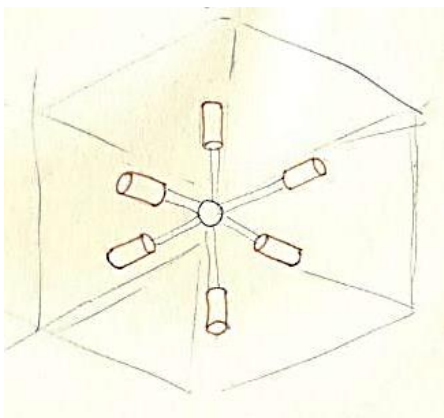
How do tilt switches work?

Tilt switches work by having a conductive object (normally a ball) that bridges 2 contacts when in a certain position.



How do we translate the tilt switches into sound?

By using electronic circuits.



By have 6 switches there will always be at least 1 that's on.

Could tilt switches work, when taking gravity and centrifugal force into account?

I don't know let's boot up Excel!

Evaluating the idea:

We know that the system will fail when centrifugal force > gravitational force.

Gravity: $F_g = m * g$

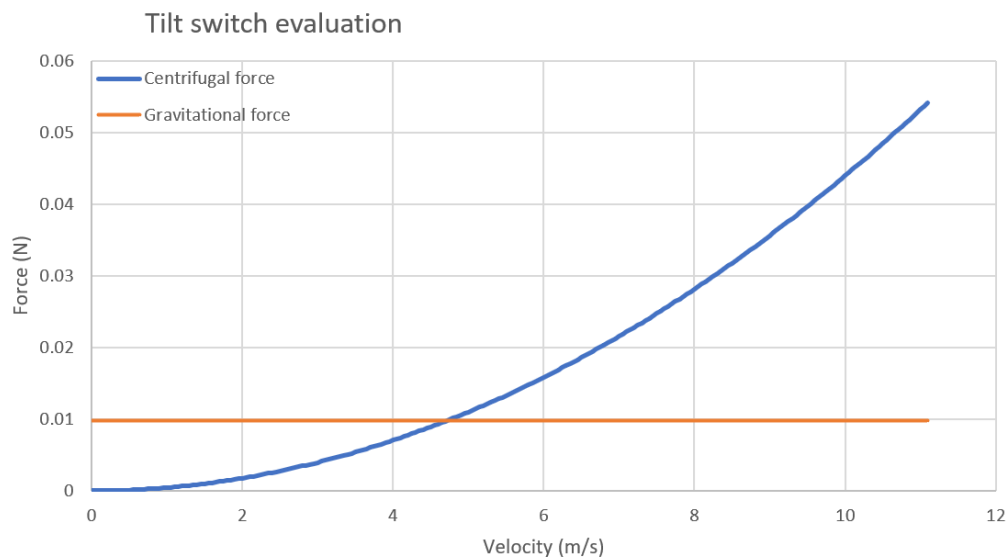
Centrifugal force: $F_c = m \frac{v^2}{r}$

We can make an automatic model in Excel. The variables are the mass of the tilt switch, the radius of the tilt switch (Centre of rotation -> tilt switch), and the radius of the outer ball.

Now we plot a graph showing the relation between the Velocity and the Forces.

Where the 2 lines intersect is the maximum velocity that the system can handle.

Centrifugal force ($F = m(v^2/r)$)		
g	9.81	m/s ²
m(tilt switch)	0.001	kg
v(bal)	0	m/s
r(bal)	0.08	m
r(sensor)	0.015	m
dv	0.05	m/s
Angular velocity	0	rad/s
V(sensor)	0	m/s
Force(sensor)	0	N
Gravity	0.00981	N



Conclusion:

For the small ball (Diameter = 16cm) the system would work up and until 17km/h.

The large ball (Diameter = 23cm) would work till 27km/h.

This is sadly too slow, especially when considering that this is not the worst scenario.

We will therefore have to abandon this idea as well.

Digital

Summary:

The same idea as was developed last year, using a digital sensor (gyroscope) and a microcontroller to determine all the factors and functions.

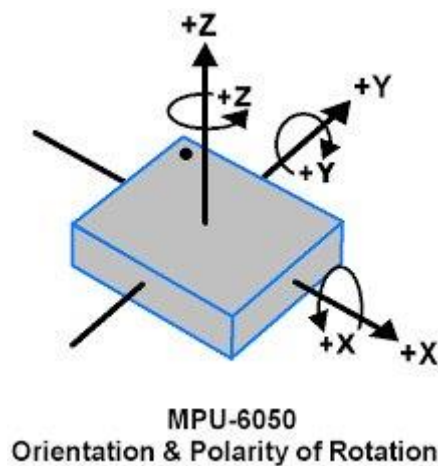
Research:

How do we measure the speed/acceleration of the ball?

By using a gyroscope, a gyroscope is a digital device that can measure change in direction and velocity which can in turn be used to determine other factors like the acceleration.

How do we transform the measurements into sound/light?

Coding, we can make functions and formulas to change parameters based on the measurements.



Evaluation:

Seeing that the goal of these first concepts is to see which ones could work and which has the most potential, making a prototype is currently outside of the scope.

We know that this system does work, it has been done by last years group and it commonly used for similar projects like drones, the UEFA soccer ball, and phones.

Conclusion

Even though analogue could work it is not desired by the client for valid reasons.

When the speeds would remain low then electronic systems could work, however the speeds will likely exceed the maximum speeds, besides it lacks the customization options desired by the client.

Therefore, the best and safest direction to go for is the digital one. It provides us with a lot of customization, a far greater speed limit, more chances for receiving outside help, and an overall more robust system.

Research

This is a bit of a research dump, every question that arises is placed here and answered.

How to Make the Sound?

The sound in our project can be generated using buzzers or speakers. They produce sound through the vibration of a diaphragm, which can be triggered by an electrical signal. By controlling the frequency and duration of the electrical signal, different sounds can be produced.

How Do Buzzers Work?

Buzzers work by converting an electrical signal into mechanical vibrations. These vibrations produce sound waves that are audible. There are two main types of buzzers: piezoelectric buzzers, which use a piezoelectric element, and magnetic buzzers, which use an electromagnet to create sound.

What Kind of Speakers Are There?

There are several types of speakers we could consider:

- **Dynamic Speakers:** These use a coil of wire and a magnet to produce sound and are common in many audio devices.
- **Piezoelectric Speakers:** Similar to buzzers, they use piezoelectric materials to generate sound and are often used in smaller devices.
- **Electrostatic Speakers:** These use a thin, electrically charged diaphragm and are known for high fidelity but are typically more expensive, complex, and less robust.

Overall, the Dynamic Speakers are cheap, light, robust, and over plenty of quality for our use.

How Do We Charge It?

Charging can be achieved using a USB-C connector, as determined by EU regulations. This ensures compatibility with common charging devices and simplifies the charging process. We can use a rechargeable lithium-ion battery, which offers a good balance between capacity and weight. Alternatively we could go for wireless charging. However, implementing a charging coil into the design will be difficult, if doable at all.

How Do We Determine Motion?

Motion can be determined using sensors such as accelerometers and gyroscopes. These sensors can measure changes in speed and direction, allowing us to track the movement of the ball. A microcontroller can process the data from these sensors to determine the ball's movement.

How Do We Control It via a Phone?

Control via a phone can be achieved using Bluetooth technology. By integrating a Bluetooth module into the ball, we can connect it to a smartphone app. The app can be used to send commands and receive data from the ball, allowing for remote control and customization.

What Colour Is Most Appropriate for the Purpose of Contrast?

High-contrast colours like bright yellow, neon green, or bright orange are typically most appropriate. These colours stand out well against various backgrounds and are easier for visually impaired individuals to detect.

What Colour Are Gym Floors?

Gym floors are often a light wood colour; they are often painted with the colours depending on the sport and the institution. Common colours include shades of blue, red, and green.

How Is the Ball Used?

The ball is used in various activities, most notably goalball. But the primary goal is to help visually impaired individuals locate it through sound. Meaning that it needs to be durable, produce consistent sound, and be easy to handle.

How Do the Materials Influence the Use?

Materials influence the ball's weight, durability, and the quality of sound produced. For instance, a lighter material might make the ball easier to handle, but it needs to be strong enough to withstand impacts. The internal materials should also ensure that the sound components are well-protected and functional.

How Do Textures Influence the Use?

Textures can affect grip and handling. A textured surface can provide better grip, making it easier to 'catch' and hold. It can also influence the ball's behaviour when it rolls, which might affect (the consistency of) the sound it produces.

Where Is the Ball Used and How Does This Impact the Project?

The ball is used in various environments, including gyms, and outdoor fields.

This impacts the project by requiring the ball to be versatile and durable in different conditions. It must be weather-resistant and capable of producing sound regardless of the environment.

Can We Create an "Analogue" Ball?

Creating an analogue ball would involve using mechanical means to produce sound, such as bells or chimes. However, based on feedback, this approach is less desirable due to limitations in sound duration and customization.

How to Prolong a Sound?

Prolonging a sound can be achieved using electronic means. Mechanically, chimes and bells can produce prolonged sounds, but they may not meet the customization needs of the project.

Chimes and Bells?

Chimes and bells can be used to produce long-lasting sounds. Chimes, especially, can continue to resonate for several seconds after being struck.

How to "Measure" or Use Tilt/Movement?

Tilt and movement can be measured using tilt switches or more advanced gyroscopic sensors. Tilt switches work by completing an electrical circuit when tilted, while gyroscopes provide more precise measurements of orientation and motion.

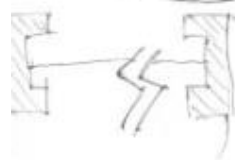
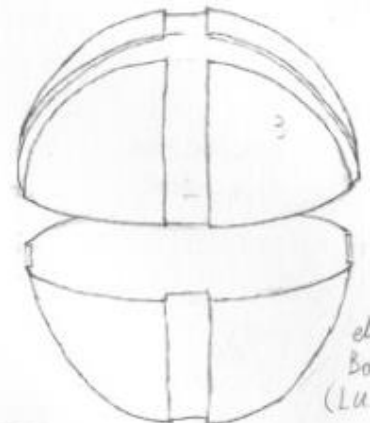
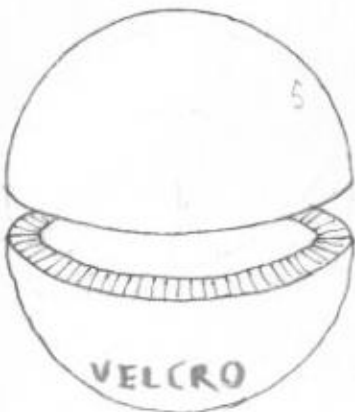
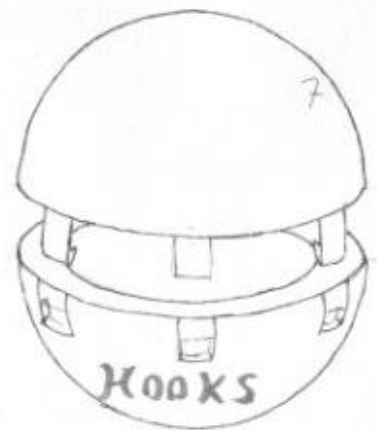
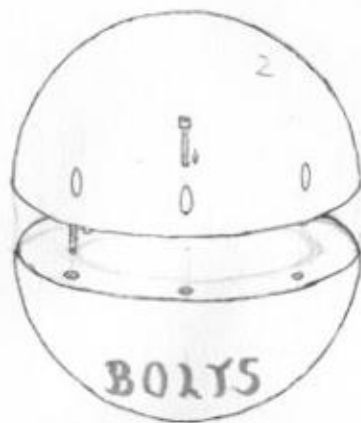
Tilt Switches – How Do They Work?

Tilt switches operate by using a small conductive object (usually a ball) inside the switch that moves and makes contact with different terminals as it tilts. This completes an electrical circuit, indicating the tilt angle.

How to Connect It to Our Phone?

Connecting the ball to a phone can be done using Bluetooth. This requires integrating a Bluetooth module into the ball's circuitry and developing a smartphone app to manage the connection and control the ball's functions.

Ideation closing mechanism



Evaluation closing

We have 10 mechanisms, and now need to decide which is best, one way of doing this is by making a Kesselring for which we need requirements.

Requirements closing mechanism

Requirement	Clarification	Importance	Category
Ease of opening	The ball should be openable without use of special tools, within a time span of x seconds.	3	Ergonomic, Usability
Elasticity	The closing mechanism should not influence the elasticity / bouncyness in less then al directions.	4	Functional
Integrity	The segments should not come appart when a force of x N is applied in the least optimal direction.	5	Functional
Understandable	The mechanism is understandable to (almost) everyone without prior knowledge.	3	Ergonomic, Usability
Cost	The connection should cost < x euro's.	2	
Lights	The mechanism allows for lights.		
Sound	The connection design does not prevent sound.		
Smoothness	No part are allowed to stick out.	5	Functional
Weight distribution	The weight of the ball can not exceed x grams, and the CG can deviate at max x mm.	4	Functional
Design freedom	The chosen connection shouldn't impact the design to much.	1	Design

Kesselring

	Ease of opening	Elasticity	Integrity	Understandable	Cost	Lights	Sound	Smoothness	Weight distribution	Design freedom	Total
Ease of opening	-	0	0	1	1	1	0	0	0	1	4
Elasticity	1	-	0	1	1	0	0	0	0	1	4
Integrity	1	1	-	1	1	1	1	0	1	1	8
Understandable	0	0	0	-	0	0	0	0	0	1	1
Cost	0	0	0	1	-	0	0	0	0	1	2
Lights	0	1	0	1	1	-	0	0	0	1	4
Sound	1	1	1	1	1	1	-	1	1	1	9
Smoothness	1	1	1	1	1	1	0	-	1	1	8
Weight distribution	1	1	0	1	1	1	0	0	-	1	6
Design freedom	0	0	0	0	0	0	0	0	0	-	0

The Kesselring shows that the most important requirements are the sound, the smoothness (safety), and the Integrity.

Idea Nr.	Scale-ing factor	1 Press fit	2 Bolts	3 Elastics	4 Screw	5 Velcro	6 Barrel	7 Hooks	8 Zipper	9 Drum	10 Hooks 2	Ideal Score
Ease of opening	4	4	4	5	5	5	4	4	4	1	2	20
Elasticity	4	5	3	4	5	3	1	3	4	4	3	20
Integrity	8	1	5	2	2	1	5	3	5	4	3	40
Understandable	1	4	4	5	5	4	3	4	5	4	1	5
Cost	2	5	3	4	3	4	3	5	4	4	1	10
Lights	4											20
Sound	9											45
Smoothness	8	5	4	3	5	5	2	1	2	3	4	40
Weight distribution	6	5	5	4	1	5	3	5	3	5	4	30
Design freedom	0	5	5	5	5	5	5	5	5	5	5	0
Total score		128	140	113	113	122	103	104	119	118	103	230
Percentage		56%	61%	49%	49%	53%	45%	45%	52%	51%	45%	100%

We decided to not count the Light and Sound requirements because we don't have enough knowledge about it to say anything about this.

Conclusion:

The best ideas are Velcro, Press fit and bolts. With bolts being the least risk out of all of them, we shall therefore go with the bolts.

Note on requirements:

This does also show the importance of good requirements, both Velcro and Press fit had the lowest integrity score, but some of the highest overall score. Such a stark difference should've been a no-go.

Prototyping

Electronic inner ball

Goals:

Safety & Durability: Design the ball to be safe and durable for the use by children, ensuring it can withstand rough handling and impacts.

Reliability: Integrate the microcontroller, gyroscope, speakers, and batteries in a way that ensures reliable performance and ease of use.

Maintenance & charging: Design the ball to be easy to maintain and charge, ensuring it remains functional over time with minimal effort. While being able to be serviced by anyone that knows how to use a screwdriver.

Documenting the process: Document the development process, testing results, and user feedback. So that I or someone else can continue later.

Research:

How do visually impaired children play with toys?

They rely on tactile and auditory feedback. Sounds should be clear and distinct.

What types of sounds are most effective for our purpose?

High-pitched, clear, and repetitive sounds are generally effective for grabbing children's attention (Is this the same for VIP?). Overly complex or startling sounds should be avoided because these will become annoying overtime.

Are there safety standard to consider?

There are safety standards for children's toys and the like. Which lines up somewhat. These are ASTM F963 and EN71.

What are the power requirements for a microcontroller handling audio and gyroscope data?

Depends on the microcontroller used. Generally, a microcontroller like the ESP-32 requires 3.3V to 5V and has a current draw of 50-250mA. 8-72Hours on an 18650 (not including the speakers).

How do gyroscopes work?

Gyroscopes measure angular velocity which can be used orientation and rotational speed of the toy. By integrating the readings over time, we can estimate linear speed. Which in turn can be used to adjust the audio feedback.

How does speaker placement affect audio in a sphere?

Placing the speakers evenly around ensures uniform sound. Making it easier for users to hear and identify the direction of sounds (Last year's feedback also mentioned this fact). Another advantage of even distribution of multiple speakers is extra customization, for example adding a doppler effect, aiding in the direction detection of the user.

Are there known best practises for designing UI for visually impaired children?

Best practices include using simple, intuitive controls, providing clear and immediate auditory feedback, incorporating tactile elements for guidance, and ensuring the interface is consistent and predictable to facilitate learning and use.

Requirements:

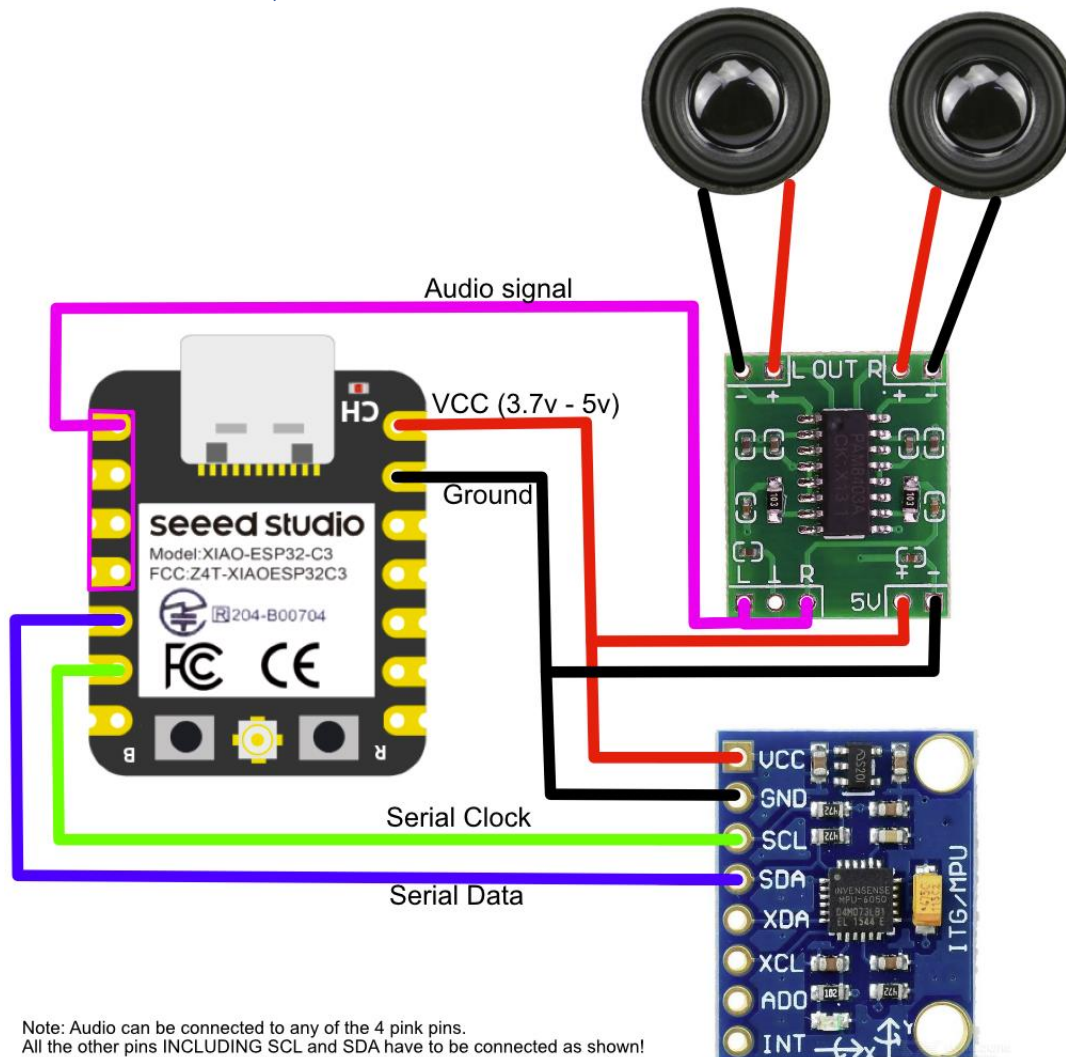
Requirement	Explanation	Achieved?	Actions
Motion detection	The ball detects movement and can determine a scale of velocity.	Y	The velocity parameters must be finetuned.
Audio feedback	The ball must produce a sound when stationary, a different sound when rolling, and the rolling sound changes with the velocity of the ball.	Y	The velocity parameters must be finetuned.
User accessibility	The audio signals must be clear and easy to distinguish, both in general and between the stationary and moving sounds.	YN	They are very easy to hear and distinguish, but they are also awful in all ways.
Gyroscope	Must be sensitive enough to detect slight changes in speed.	Y	
Speakers	The speakers must produce clear and loud sounds in a 360° audio coverage.	Y	Are 6 speakers necessary or could we do with 4?
Batteries	Provide enough power to run all the electronics. Must be rechargeable.	Y	We do not know how long the batteries last.
Shell	The shell must be durable enough to withstand regular use and impacts from being dropped/thrown. It should also house all the electronics.	Y	The shell is currently only 3D-printable. Will this stay this way or does this need to change?
Sound	The prototype should have a maximum dB of at least 80dB	Y The minimum recorded was 84.7!	Make the volume adjustable.

Proof of principle

Goals:

- Get an understanding of both the hardware and software.
- Create a working code, and finetune it somewhat.
- Create a list of what is needed for an actual prototype.

Electric hardware set-up:



Code:

Please see the Appendix for the code

Feedback and actions:

- **The speeding up sound isn't continues, you can hear the steps very clearly.**
I'll investigate the code.
Update: fixed, some scaling issue caused there to be 5 steps, now there are ~500.
- **The speakers are getting warm, will this be an issue?**
This is mainly because of the loudness and the sound, when toning it down the issue will be less, but we should test this.
- **In one orientation the sound doesn't work properly anymore.**
I'll investigate the issue.
Update: Fixed, the z-value of the gyroscope is very inaccurate and can therefore not be used, changed the code to only look at the change in angle.

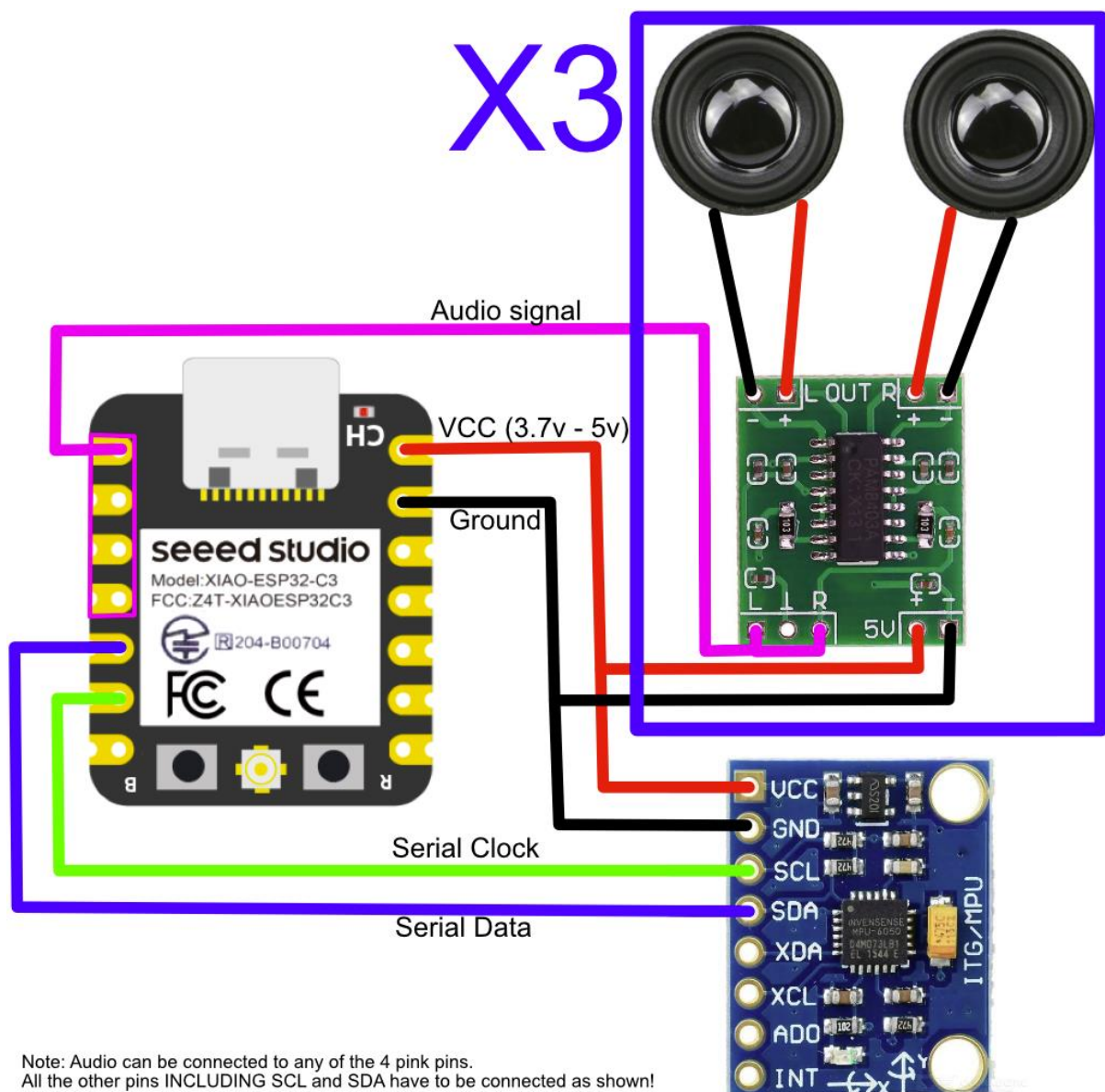
Prototype

Goals

- Create a full inner ball with both the hardware, software, and electronics working together.
- Get an understanding of what is needed or asked of the outer foam ball.

Electronics

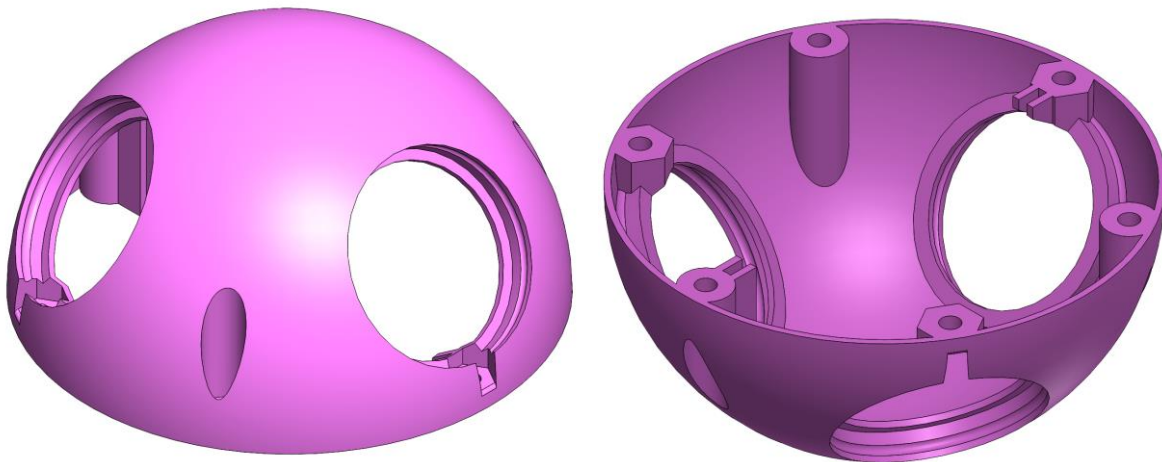
The electronics are the same as in the proof of principle, except for 3 times more amplifiers and speaker.



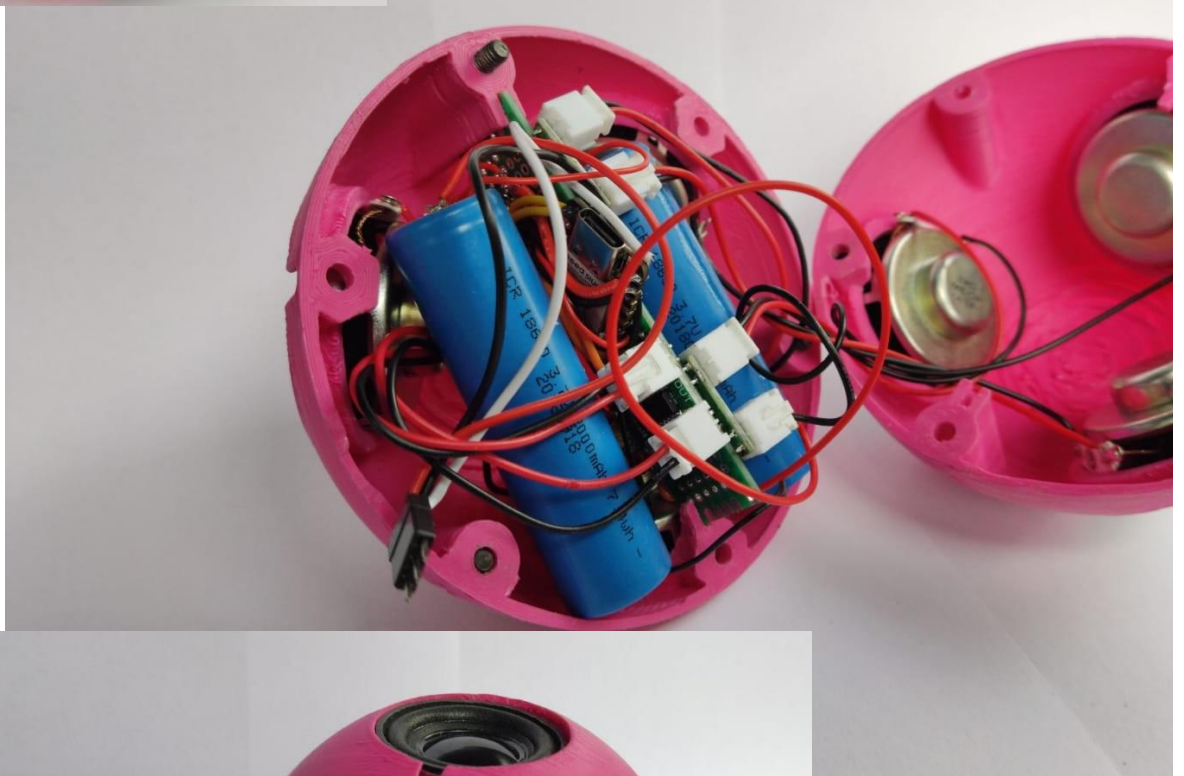
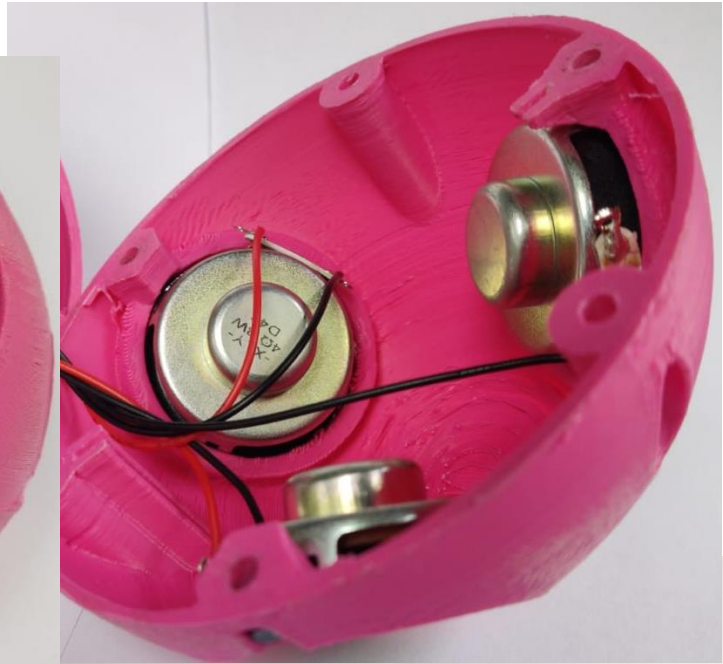
Shell

The plastic housing for all the electronics including the speakers.

Requirements	Explanation	Achieved?	Actions
Robust	The shell is durable enough to withstand rough play, and harder impact associated with this.	?	Not enough test conducted.
Weight	The entirety of the prototype weighs <400g	Y It only weighs 249Grams!	
Component housing	The shell must have compartments to securely house the microcontroller, gyroscope, amplifiers, speakers, and batteries.	N	There are compartments for everything except the batteries.
Surround sound	The speakers are housing in a manner that allows them to provide (close to) 360° sound.	Y	Again, are 6 speakers needed or could we do with less?
Assembly	The shell can be easily assembled and disassembled without the need of force.	Y	
Dimensions	The ball has a maximum outer diameter of 120mm, With preferred diameter of 100mm.	Y	



Complete



Feedback

- ITS WAY TO LOUD!
- Batteries don't let them float around like that.
- The sound needs to change.
- That is beautiful, it looks like art!
- Why 6 speakers?
- I like the closing mechanism.
- How will it be produced? Because as it is currently only 3D-printing is feasible.

Plan of improvement

First prototype outer ball

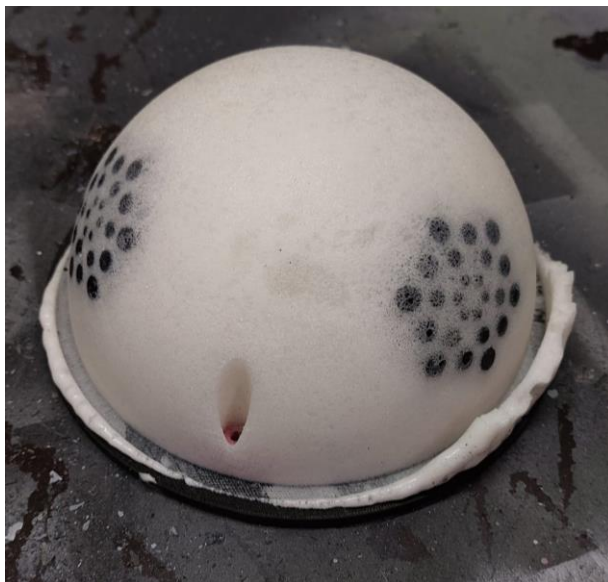
Main goals:

- Is in house foam moulding for low quantities feasible, reasonable, and logical?
- Understanding and experiencing flexible foam casting.

Requirements design

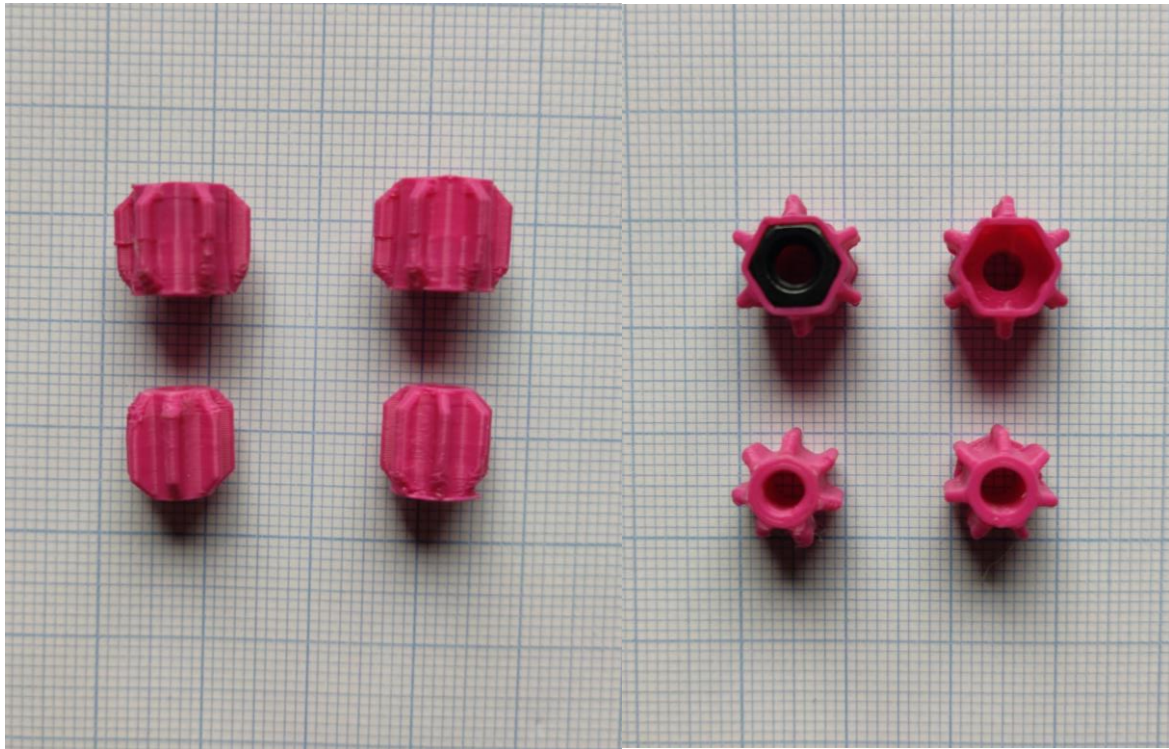
Requirement	Explanation	Achieved?	Actions
Mouldable	Mouldable in a 2 part mould.	YN It is doable but not easy to demould.	Using a silicone mould, increasing the draft angles, or making it >2 part mould.
Connection	Uses nuts and bolts as connection. (determined at ...)	Y Connections worked perfectly,	
Inserts	To achieve a strong and reliable connection uses moulded in inserts. (See insert requirements)	Y- The inserts worked perfectly, but difficult in the mould.	Redesign the mould to make the inserts easier to add. (Might require a slight redesign of the inserts.)
Sound	The design is not allowed to dampen the sound more than	Y The outer_ball does not dampen the sound.	
Egality of softness	The ball has the same softness in all places, main places of concern are the speakers and the inserts at the seam.	N The softness is different nearly everywhere. This is caused by both the design and the moulding.	Make the wall thickness equal wherever possible, thicken the speaker areas, and move the inserts further towards the centre.

Note on requirements: As you can see, we do not value a lot off the earlier requirements this is to simplify the first prototype and make it into a prove of principle.



InsertsV1

I designed in-mould inserts based on the inserts that I had seen for injection and rotation moulding. *Technical drawings can be found in the Appendix.*



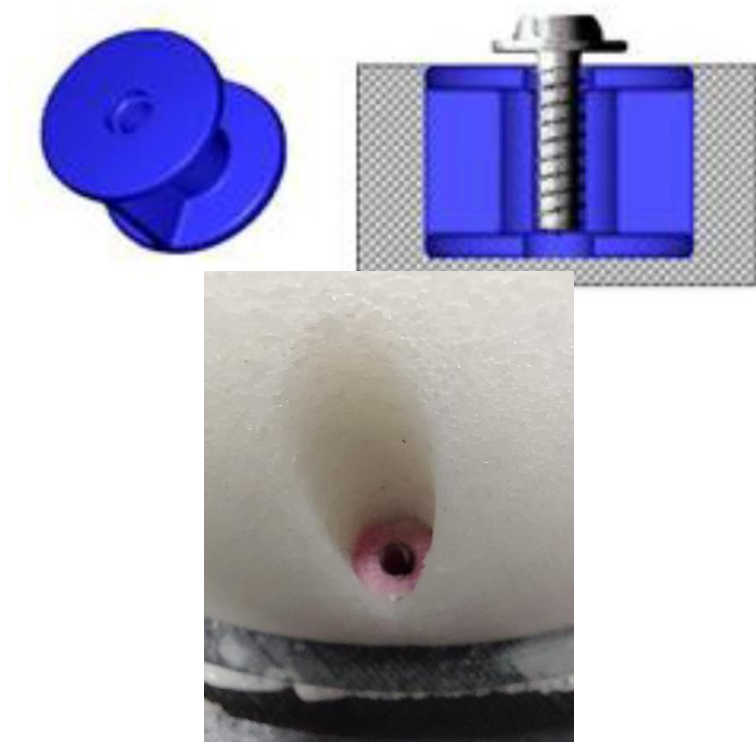
Feedback:

After testing out the foam and experiencing that it doesn't stick that well to 3d-printed plastic. We concluded that these inserts would likely not suffice and were likely to 'pull out' during use. What we needed was requirements.

Research:

Even though foam inserts exist not a lot of information can be found.

What we do find is that they have large surface areas, perforations, and big 'fins' to prevent rotation.



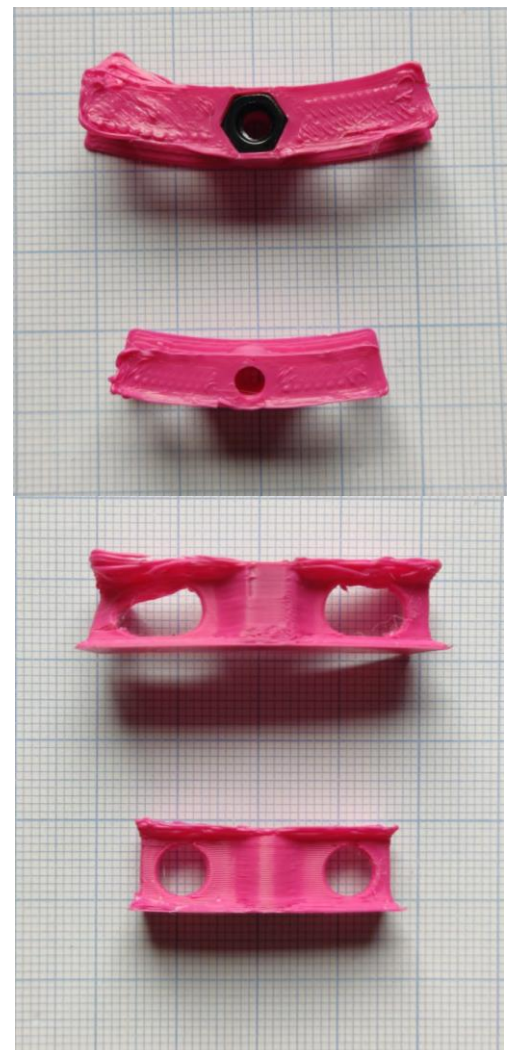
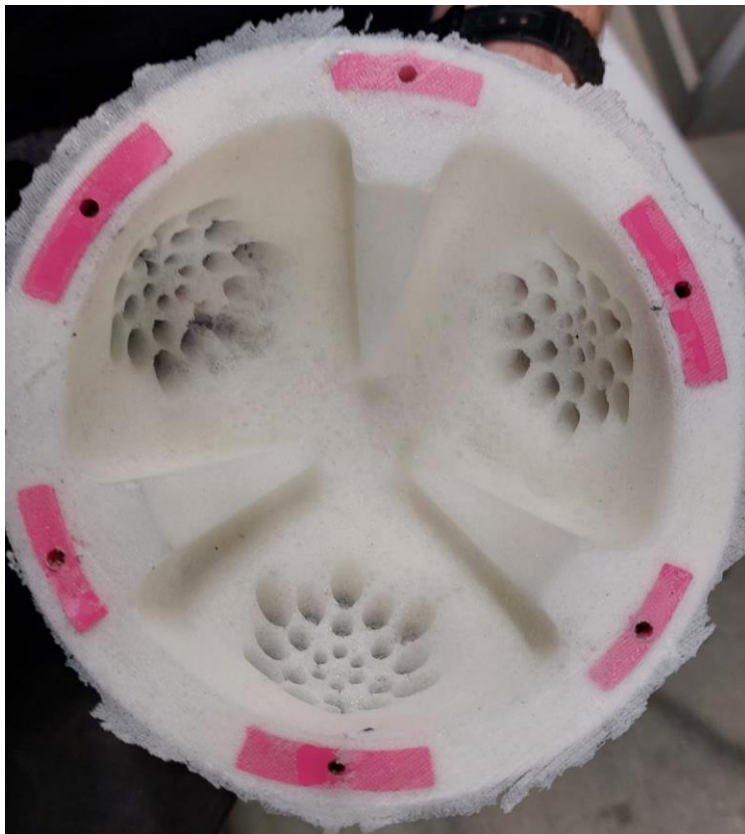
Requirements inserts

Requirement	Explanation	Achieved	Actions
Prevents foam crushing	When the ball is connected the foam can't be compressed at the connection points.	Y	
Properly attached	The inserts must be properly attached, this means they can't fall out, or be pulled out during reasonable use.	Y	
No twisting	The inserts can't twist when both halves are connected.	YN They can twist slightly when connecting the 2 halves.	Denser foam might solve this, otherwise designing the inserts to interlock could solve it.
Safety	The inserts must be sufficiently hidden; you are not allowed to "feel" where the inserts are due to hardness.	YN You can feel them, this is more a problem with the outer_ball shape.	-No action needed for now.
Connection	One of the 2 different inserts must accommodate either a nut or have threads itself so that the connection can work.	Y	

InsertsV2

Using the new requirements and the research conclusions I designed new inserts. These were deemed good enough for the first moulding tests.

Technical drawings can be found in the Appendix.



Feedback proto-1:

Mould:

- Alignment must be done manually by eye.
- Inserts must be temporarily glued (this is because the mould was designed for different inserts)
- Screws couldn't handle the force, causing the mould to open during the moulding.

Design:

- The seem is very noticeable, this can be hidden in plain sight by the design.
- It's very plain.
- The thickness in the speaker area is too thin.
- Wall thickness is too small in areas causing both the inserts to be too close.
- Wall thickness is not equal everywhere, this is more noticeable than expected (different foam might make this unimportant).
- The electronics ball can rotate, it should be fixed some way or another.
- The outside is open foam and thus a dust magnet, how can this be solved?

Inserts:

- + The inserts work perfectly as intended.
- They can be seen and felt from the outside.
- They might affect the feel of the ball around the seem (More tests/ 2nd proto are needed for certainty).
- + Good thing that we redesigned them the first ones would likely not have worked
- They do twist slightly, interlocking might be a good idea.

Loose Feedback:

- Requirements for the mould would've been handy :)



Prototype 2

Note: Green in requirements list means; changed or added requirement.

Main goals:

- Create a feasible design.
- Create a design that looks good, and functions as intended.
- User interface.

Requirements design

Requirement	Explanation	Achieved?	Actions
Mouldable	Mouldable in a 2 part mould.	Nah	
Connection	Uses nuts and bolts as connection. (determined at ...)	-	
Inserts	To achieve a strong and reliable connection uses moulded in inserts. (See insert requirements)	-	
Sound	The design is not allowed to dampen the sound more than	-	
Egality of softness	The ball has the same softness in all places, main places of concern are the speakers and the inserts at the seam.	-	
Hidden seem	The seem is included or at lease accounted for in the design of the ball.	-	
User Interface	The ball has a user interface consisting of a on/off button and a usb-C port these parts do not need to share the same interface. Moreover the interface is not allowed to stick out due to safety.		
Speaker holes	The softness and thickness of the ball does not change significantly between the speaker holes and the rest of the ball.	?	
Centre of mass.	The centre off mass can deviate ...mm from the centre of the sphere.	-	
Internal ball	The electronic ball does not shift (relative to the outer ball) during the use of the ball.	?	
Symmetry	The 2 halves are symmetrical and can use the same mould (or with an insert blocking a part of the mould).	-	
Texture	An outer texture that is nice to touch or hold (talk to the other groups for their research).		

Requirements mould

Requirements	Explanation	Achieved?	Actions
2 Part mould	The mould is a 2-part mould, remember the foam is flexible so draft does not need to be positive.		
Alignment	Core and cavity can easily be aligned.		
Closing	The mould can be closed with a force of In a time of....		
Insert alignment	The mould has something in place to allow the easy alignment of the inserts. (Also see requirements inserts.)		
Reuse	The mould can be used non-destructively, allowing re-use of the mould.		

Requirements inserts

Requirement	Explanation	Achieved	Actions
Prevents foam crushing	When the ball is connected the foam can't be compressed at the connection points.		
Properly attached	The inserts must be properly attached, this means they can't fall out, or be pulled out during reasonable use.		
No twisting	The inserts can't twist in relation to each other when both halves are connected.		
Safety	The inserts must be sufficiently hidden; you are not allowed to "feel" where the inserts are due to hardness.		
Connection	One of the 2 different inserts must accommodate either a nut or have threads itself so that the connection can work.		
Interlocking	The inserts lock together, completely preventing twisting. Added benefit is alignment within the mould.		
Foam lock	The foam is self-skinning, meaning that we need holes or other features to really lock the inserts in place.		

Sketches inserts

Appendix

Autopsy components list

Made by group 3

Speaker

Weight: 17 grams
Diameter: 40 mm
Height: 17 grams
Information Z-D43, 3W



Rechargeable battery + converter

Weight: 50 grams
Length: 70 mm
Width: 26,5 mm
Height: 27,5 mm
Information: Bat nr. ICR 18650, 20180918



Arduino nano

Weight: 5 grams
Length: 45 mm
Width: 18,5 mm



Battery 9V

Weight: 33,9 grams
Length: 26,5 mm
Width: 17,5 mm
Height: 48,5 mm



Sd Card reader

Weight: 4 grams
Length: 46 mm
Width 24 mm
Information: Hw-125



Power switch

Weight: <1 gram
Dimensions: 14x7x30 mm



Gyroscope

Weight: 2 grams
Length: 20,5
Width: 11
Information: Gy-291



Motherboard + stands

Weight: 51 grams
Diameter: 80 mm
Height: 37 mm

Amplifier

Weight: 2 grams
Length: 21,5 mm
Width: 19 mm
Information: Hw-104



1/2 Core dome

Weight: 54 grams
Diameter: 90 mm
Height: 60 mm

Functional tests

These tests were conducted with most people present; document made by Saskia Honings.

Test Visioball 06-03-2024

Straightness of roll

Over a distance of 12 meter the ball goes of ... mm

Fast

1. Straight with the tape line
2. ?
3. ?

Slow

1. 700 mm to the left
2. 150 mm to the left
3. Straight with the tape line

DB Test

Case 1, 2 and 3 the phone is 1m from the object. We measure on the outside of the ball.

C1: Stationary speakers to the side

	Phone 1 (iPhone)	Phone 2 (Android)	Phone 3 (Android)	Phone 4 (Android)
Average	58.4 DB	61.1 DB	41.5 DB	39.0 DB
Maximum	66.5 DB	69.1 DB	55.6 DB	62.1 DB

C2: Rolling speakers to the side

	Phone 1 (iPhone)	Phone 2 (Android)	Phone 3 (Android)	Phone 4 (Android)
Average	65.9 DB	65.0 DB	38.4 DB	40.8 DB
Maximum	72.9 DB	74.1 DB	75.5 DB	76.6 DB

C3: Stationary speakers facing the phones

	Phone 1 (iPhone)	Phone 2 (Android)	Phone 3 (Android)	Phone 4 (Android)
Average	55.3 DB	60.0 DB	39.4 DB	37.0 DB
Maximum	62.6 DB	68.2 DB	55.7 DB	52.8 DB

Consistency of sound

Rolling the ball on a distance of 12 meters and timing how long it does over those meters. Here from we can calculate the speed when there is less feedback from the ball.

C1: Floor at the workshops

	Distance	seconds	AVG sec	Ticks in one throw	Breaks in one throw
1	12 meter	1.99	1.99		
2	12 meter	1.82, 1.87	1.845		
3	12 meter	2.02, 1.97	1.995		
4	12 meter	1.63, 1.8, 1.66	1.697	3	
5	12 meter	1.46, 1.55	1.505		
6	12 meter	1.38, 1.26, 1.28	1.307	4	
7	12 meter	1.08, (0.82)	1.040	3	

8	12 meter	2.78, 2.39	2.585		2-3
9	12 meter	2.45, 2.51	2.480		3-4
10	12 meter	2.35	2.35		1-2
11	12 meter	2.43, 2.72	2.575	3	

C2: Floor at IDE

	Distance	seconds	AVG sec	Ticks in one throw	Breaks in one throw
1	6 meter	2.11		6	2
2	6 meter	1.53		3	2
3	6 meter	1.62			0
4	6 meter	1.53			0
5	6 meter	1.79			1

Speaker straight with the tape line

	Distance	seconds	AVG sec	Ticks in one throw	Breaks in one throw
1	6 meter	1.93		2	
2	6 meter	1.77		6	4

On/off with the tape line

	Distance	seconds	AVG sec	Ticks in one throw	Breaks in one throw
1	6 meter	1.94		2	1-2
2	6 meter	1.81			1

Final code prototype

Instructions:

To run this code you'll need the hardware, you'll also need to have the Adafruit_MPU6050 library installed.

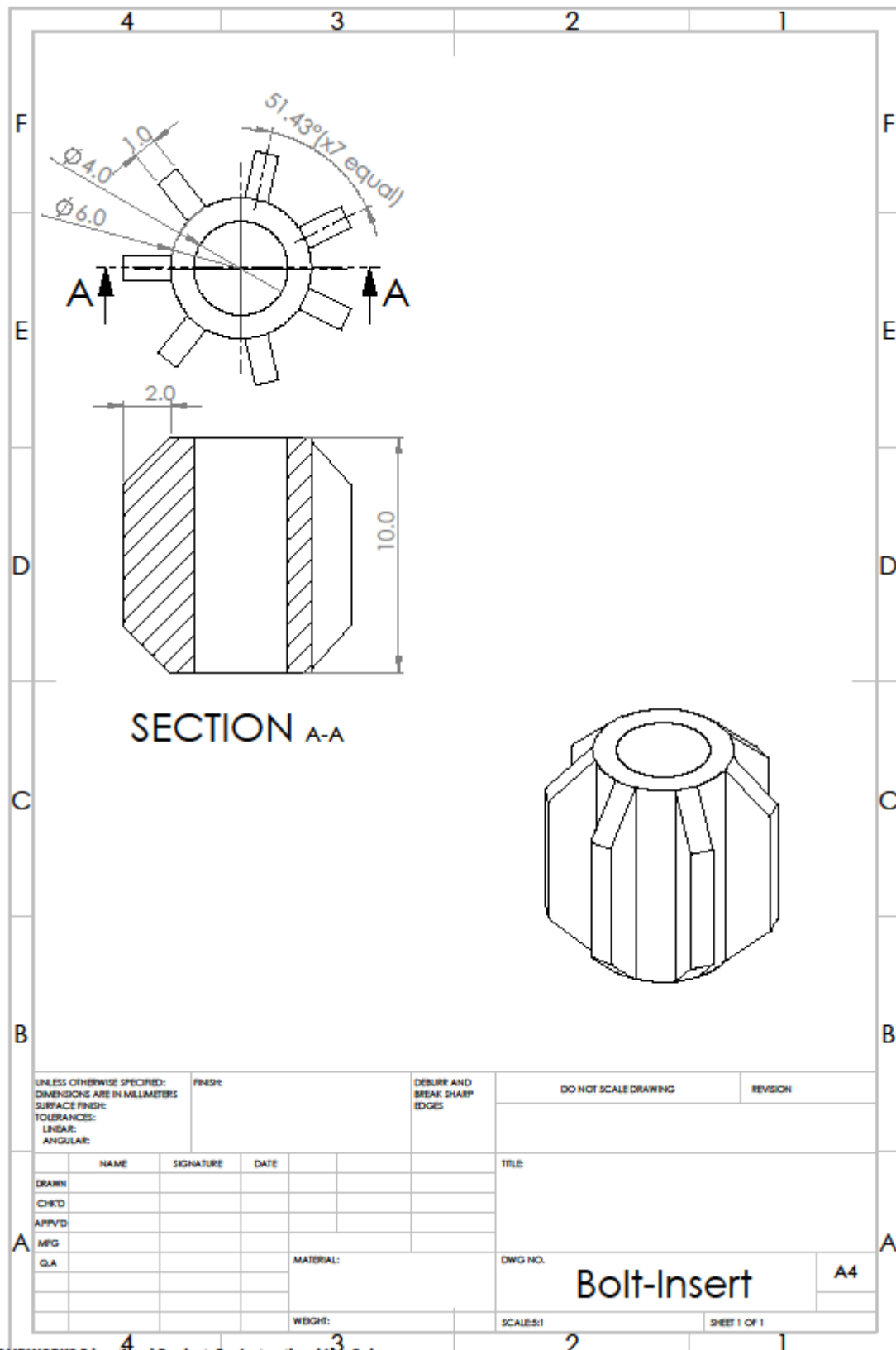
```
1. #include <Wire.h>
2. #include <Adafruit_MPU6050.h>
3. #include <Adafruit_Sensor.h>
4. #include <Arduino.h>
5.
6. Adafruit_MPU6050 mpu;
7.
8. const int PWM_PIN = 2; // Choose a GPIO pin capable of PWM (e.g., GPIO 13)
9. const int BEEP_FREQUENCY = 50; // Frequency for stationary beep in Hz
10. const int BUZZ_MIN_FREQUENCY = 100; // Minimum frequency for buzzing in Hz
11. const int BUZZ_MAX_FREQUENCY = 1000; // Maximum frequency for buzzing in Hz
12. const int CHANNEL = 5; // PWM channel
13. const int RESOLUTION = 8; // 8-bit resolution
14. const float STATIONARY_THRESHOLD_GYRO = 1.0; // Threshold for detecting stationary state (gyroscope)
15.
16. // User-defined variables to control frequency scaling
17. const float GYRO_SCALE_FACTOR = 0.2; // Scaling factor to adjust the rate at which the frequency increases
18.
19. void setup() {
20.     // Initialize the serial communication
21.     Serial.begin(115200);
22.
23.     // Initialize the MPU6050
24.     if (!mpu.begin()) {
25.         Serial.println("Failed to find MPU6050 chip");
26.         while (1) {
27.             delay(10);
28.         }
29.     }
30.
31.     mpu.setAccelerometerRange(MPU6050_RANGE_2_G);
32.     mpu.setGyroRange(MPU6050_RANGE_250_DEG);
33.     mpu.setFilterBandwidth(MPU6050_BAND_21_HZ);
34.
35.     Serial.println("MPU6050 ready!");
36.
37.     // Configure the PWM functionalities
38.     ledcSetup(CHANNEL, BEEP_FREQUENCY, RESOLUTION);
39.     ledcAttachPin(PWM_PIN, CHANNEL);
40. }
41.
42. void loop() {
43.     sensors_event_t a, g, temp;
44.     mpu.getEvent(&a, &g, &temp);
45.
46.     // Calculate the magnitude of the gyroscope vector
47.     float gyroMagnitude = sqrt(g.gyro.x * g.gyro.x + g.gyro.y * g.gyro.y + g.gyro.z * g.gyro.z);
48.
49.     // Apply scaling factor to delay the maxing out of frequency
50.     float scaledGyroMagnitude = gyroMagnitude * GYRO_SCALE_FACTOR;
51.
52.     // Check if the ball is stationary
53.     if (gyroMagnitude < STATIONARY_THRESHOLD_GYRO) {
54.         // Stationary - Beep sound
55.         ledcWriteTone(CHANNEL, BEEP_FREQUENCY);
56.         ledcWrite(CHANNEL, 128); // 50% duty cycle
57.         //Serial.println("Stationary");
58.     } else {
59.         // Moving - Buzz sound with frequency proportional to speed
```

```
60.     int buzzFrequency = map(scaledGyroMagnitude, 0, 10, BUZZ_MIN_FREQUENCY,
BUZZ_MAX_FREQUENCY);
61.     buzzFrequency = constrain(buzzFrequency, BUZZ_MIN_FREQUENCY, BUZZ_MAX_FREQUENCY); //
Constrain to avoid exceeding limits
62.     ledcWriteTone(CHANNEL, buzzFrequency);
63.     ledcWrite(CHANNEL, 128); // 50% duty cycle
64.     //Serial.print("Moving - Frequency: ");
65.     Serial.println(buzzFrequency);
66. }
67.
68. delay(100);
69. }
70.
```

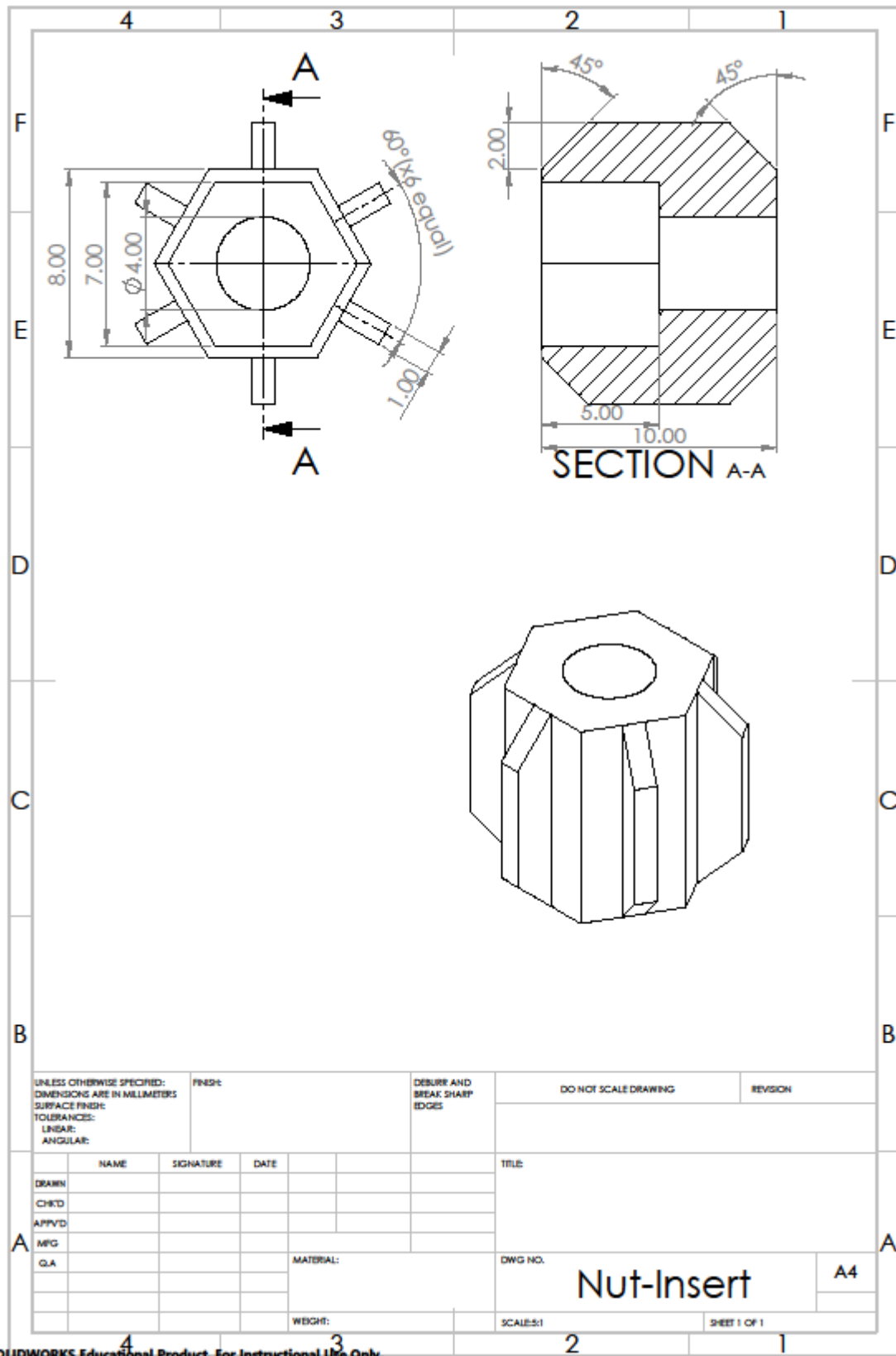
Technical Drawings

Due to limitations in Word the quality is lacking, for better quality see separately uploaded files.

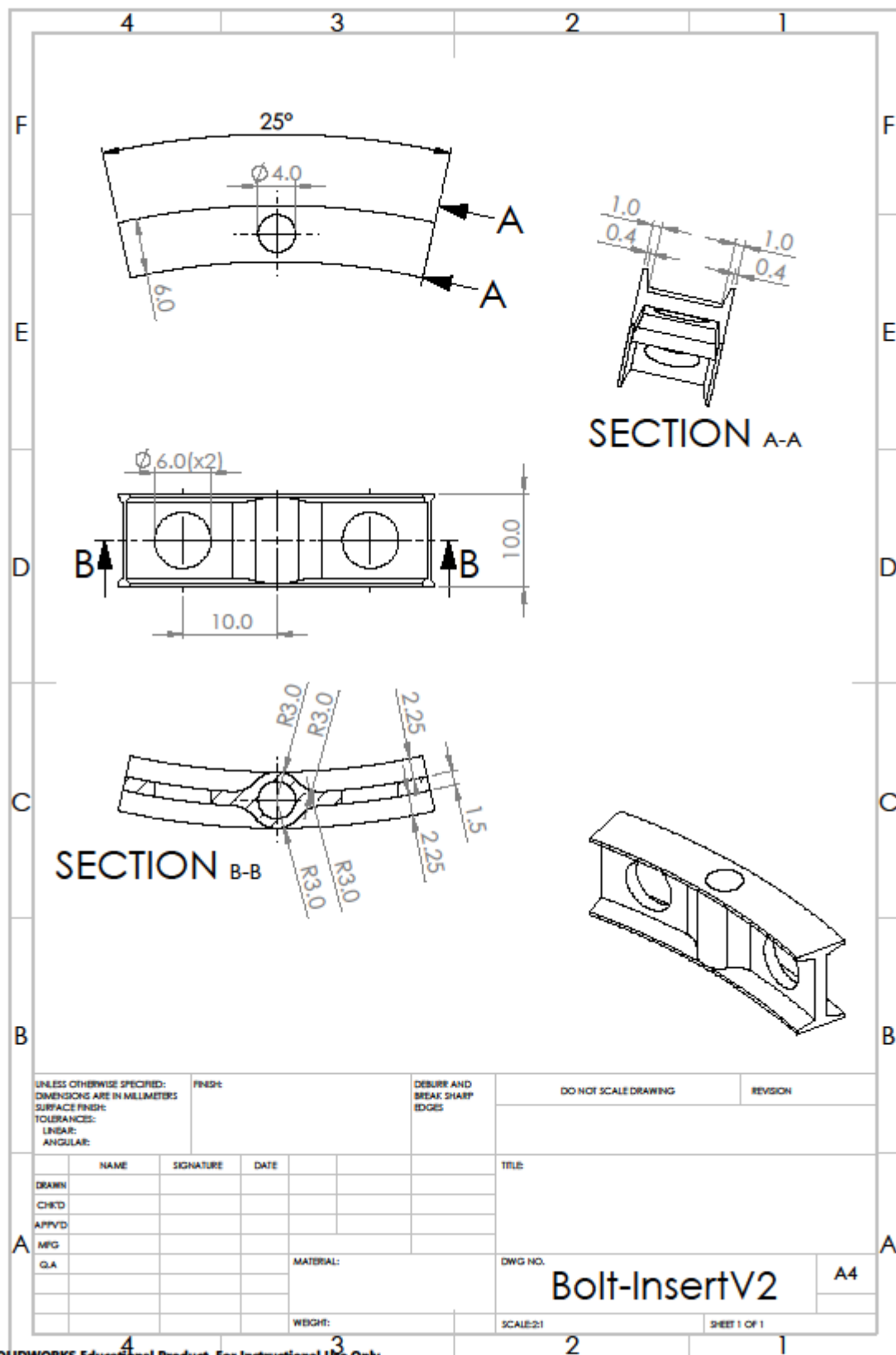
Bolt insert:



Nut insert:



Bolt insert V2:



Nut insert V2:

